Python Program Analysis & Documentation

Technical Report • Comprehensive Script Execution & Output Logs

1. Travel Budget Report (Program 5)

Calculates multi-currency expenses, conversion fees, total costs, and validates against a predefined travel budget.

Source Logic Snippet

```
categories = ["Accommodation", "Food", "T
ransport", "Shopping"] # Processing expen
ses with conversion & fee total_expenses
= 0 # Budget validation and reporting
```

Execution Output Log

```
Category: Accommodation
Original: 500 USD | Converted: Rs. 42500
Fee: Rs. 850.0 | Total: Rs. 43350.0

Category: Food
Original: 200 EUR | Converted: Rs. 18400
Fee: Rs. 368.0 | Total: Rs. 18768.0

Total expenses: Rs. 86343.0
Status: Within budget (Rs. 13657.0
remaining)
```

2. ATM Transaction System (Program 7)

Simulates interactive banking transactions including deposits, withdrawals with limits, balance validations, and summary reporting.

Execution Output & Summary

```
===== ATM TRANSACTION SUMMARY =====
Final Balance: Rs. 3500.0
Total Transactions: 3
- Deposit: Rs. 1500.00
- Withdrawal limit exceeded
- Withdrawal: Rs. 3000.00
```

3. Student Academic Performance Analysis (Program 8)

Processes nested dictionaries containing student subject marks, computes averages, tracks improvement/decline, and categorizes performance.

Student: Ravi

```
Maths - Average: 70.0 (Improvement: 20)
Python - Average: 45.0 (Decline: 10, Weak)
English - Average: 75.0 (Improvement: 10)
Overall Average: 63.33 • Good
```

Student: Sita

```
Maths - Average: 85.0 (Improvement: 10)
Python - Average: 80.0 (Improvement: 10)
English - Average: 90.0 (Improvement: 10)
Overall Average: 85.0 • Excellent
```

4. Shopping Bill & Budget Adjuster (Program 9)

Computes bill totals, applies tier-based discounts (10% for orders ≥ 1000), and dynamically adjusts quantities if budget is exceeded.

Shopping Bill Execution Log

```
Shopping Bill
Book Amount: 300
Pen Amount: 100
Bag Amount: 1000
Total: 1400 | Discount: 140.0
Final Amount: 1260.0 (Budget adjusted successfully)
```